INCH-POUND
MIL-DTL-32620
31 October 2018
SUPERSEDING
MC/PD 05 2011 SYSCOM
21 September 2011

DETAIL SPECIFICATION

UNDERSHIRT, FLAME RESISTANT, LIGHTWEIGHT, COLD WEATHER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a flame resistant (FR), lightweight, cold weather undershirt. The undershirt will be one type and in the following sizes, as specified (see 6.2).

X-Small Small Medium Large X-Large XX-Large XXX-Large

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

COMMERCIAL ITEM DESCRIPTIONS

A-A-55217 - Thread, Aramid, Spun Staple

DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-DTL-32075 - Label: For Clothing, Equipage, and Tentage, (General Use)

(Copies of these documents are available online at https://quicksearch.dla.mil/.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN ASSOCIATION OF TEXTILE CHEMISTS AND COLORISTS (AATCC)

AATCC EP9 - Visual Assessment of Color Difference of Textiles

AATCC TM8 - Colorfastness to Crocking: Crockmeter Method

Comments, suggestions, or questions on this document should be addressed to Marine Corps Systems Command, 2200 Lester Street, Quantico, VA 22134 ATTN: SEAL-SE-STDS or emailed to USMC_STDZ@usmc.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

AMSC N/A FSC 8415

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AATCC TM16.3	-	Colorfastness to Light: Xenon-Arc
AATCC TM20A	-	Fiber Analysis: Quantitative
AATCC TM61	-	Colorfastness to Laundering: Accelerated
AATCC TM70	-	Water Repellency-Tumble Jar Dynamic Absorption Test
AATCC TM135	-	Dimensional Changes of Fabrics after Home Laundering

(Copies of these documents are available online at https://www.aatcc.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ANSI/ASQ Z1.4	-	Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available online at www.asq.org.)

ASTM INTERNATIONAL

ASTM D737	-	Standard Test Method for Air Permeability of Textile Fabrics
ASTM D1776/D1776M	-	Standard Practice for Conditioning and Testing Textiles
ASTM D2594	-	Standard Test Method for Stretch Properties of Knitted Fabrics Having Low Power
ASTM D3512/D3512M	-	Standard Test Method for Pilling Resistance and Other Related Surface Changes of Textile Fabrics: Random Tumble Pilling Tester
ASTM D3776/D3776M	-	Standard Test Methods for Mass Per Unit Area (Weight) of Fabric
ASTM D3787	-	Standard Test Method for Bursting Strength of Textiles - Constant-Rate-of-Traverse (CRT) Ball Burst Test
ASTM D6193	-	Standard Practice for Stitches and Seams
ASTM D6413/D6413M	-	Standard Test Method for Flame Resistance of Textiles (Vertical Test)
ASTM E96/E96M	-	Standard Test Methods for Water Vapor Transmission of Materials

(Copies of these documents are available online at www.astm.org.)

CRC PRESS

Hayes, A. Wallace (editor) and Kruger, Claire L. (editor). *Hayes' Principles and Methods of Toxicology*. 1989. 394-396.

(There are a number of ways through which this book may be obtained, such as online retailers, bookstores, public library, to name a few.)

JOHN WILEY & SONS INC

Marzulli, Francis and Maibach, Howard. "Contact Allergy: Predictive Testing in Humans." *Advances in Modern Toxicology: Dermatoxicology and Pharmacology, Volume 4*. John Wiley & Sons Inc. 1977. 353-372.

(Copies of this document are available from the U.S. Army Center for Health Promotion and Preventative Medicine, ATTN: MCHB-DC-TTE, Bldg., E-2100, Aberdeen Proving Ground, MD 21010-5422.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 Inspections.

3.1.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.1.2 Conformance inspection. When specified (see 6.2), a sample shall be subjected to a conformance inspection in accordance with 4.3.

3.2 Recycled, recovered, environmentally preferable, or biobased materials. Recycled, recovered, environmentally preferable, or biobased materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.3 Materials and components.

3.3.1 Materials.

3.3.1.1 Basic material. The basic material for the undershirt shall be a flame resistant, circular knit jersey fabric constructed with a modacrylic/FR rayon blend, Longworth style 1922, or equal. The circular knit construction shall provide multi-dimensional moisture management. The color of fabric shall be Coyote 498 matching the reference standard furnished by the Government (see 6.2). The fabric shall show no toxicity when used as intended (see 6.1) and when tested as specified in 4.4.5. The fabric shall conform to the physical requirements specified in table I when tested as specified in paragraph 4.4.1.

TABLE I. Material physical requirements.

Material Characteristic	Material Requirement	
	Basic Material	**Cuff Material**
Construction	Jersey circular knit	Circular, 1x1 rib knit
Fiber content	Modacrylic/FR rayon blend	Modacrylic/FR rayon blend
Weight	4.7±0.4 oz./yd^2	4.7±0.4 oz./yd^2
Colorfastness (minimum)		
Laundering (3 cycles)	3.0	3.0
Crocking		
Wet	4.0	4.0
Dry	4.0	4.0
Light, Xenon	4.0	4.0
Dimensional stability (3 cycles) (maximum)		
Wales	5.0%	5.0%
Courses	5.0%	5.0%
Bursting strength (minimum)	60 pounds	N/A
Pilling (minimum)	3.0	N/A
Visual shade matching		
Daylight D75 (7500±200K)	Good match to standard sample	Good match to standard sample
Incandescent (2856±200K)	Good match to standard sample	Good match to standard sample
Air permeability (minimum)	200 cubic feet/square feet/minute	N/A

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TABLE I. Material physical requirements – Continued.

Material Characteristic	Material Requirement	
	Basic Material	Cuff Material
Fabric stretch (length) (minimum)	30%	30%
Fabric growth (length) (maximum)	5%	5%
Water sorption and wicking		
Back (water sorption)	<5 seconds	N/A
Face (wicking) (minimum)	0.4 inches/10 seconds	N/A
Moisture Vapor Transmission Rate (MVTR) (minimum)	1400 grams/square meter/24 hours	N/A
Moisture wicking (maximum)		
Wales	6 inches in 35 minutes	6 inches in 35 minutes
Courses	6 inches in 35 minutes	6 inches in 35 minutes
Drying time (maximum)	45 minutes	45 minutes
Vertical flame (initial) Wales and courses (maximum)		
After flame	2.0 seconds	2.0 seconds
After glow	5.0 seconds	5.0 seconds
Char length	5.0 inches	5.0 inches
Melt/drip	No melt/no drip	No melt/no drip
Vertical flame (after 5 launderings) Wales and courses (maximum)		
After flame	2.0 seconds	2.0 seconds
After glow	5.0 seconds	5.0 seconds
Char length	5.0 inches	5.0 inches
Melt/drip	No melt/no drip	No melt/no drip

3.3.1.2 Cuff material. The basic material for the undershirt collar and cuffs shall be an FR, circular 1x1 rib knit jersey fabric constructed with a modacrylic/FR rayon blend, Longworth style 2008, or equal. The color of fabric shall be Coyote 498 matching the reference standard furnished by the Government (see 6.2). The fabric shall show no toxicity when used as intended (see 6.1) and when tested as specified in 4.4.5. The fabric shall conform to the physical requirements specified in table I when tested as specified in paragraph 4.4.1.

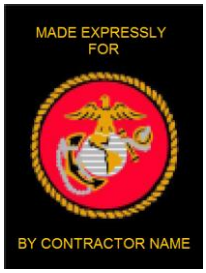
3.3.2 Components.

3.3.2.1 Thread. The thread for the undershirt shall be an aramid, Tex size 27 conforming to A-A-55217, type II. The color of the thread shall be Coyote 498.

3.3.2.2 Labels.

3.3.2.2.1 United States Marine Corps (USMC) exclusive label. Each undershirt shall have a heat transfer label, 33 millimeters x 27 millimeters (1.299 inches x 1.063 inches) in size adhered to the inside of the undershirt at the center back neck ¾±¼ inch below the collar joining seam. Placement of the Marine Corps exclusive label shall be to the left (as worn) of the combination label. The information on the label shall include the following: "Made Expressly for" at the top, followed by a USMC emblem in the center and the contractor's name below the emblem in the configuration below:

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3.3.2.2.2 Flame resistant organizational gear (FROG) label. Each undershirt shall have a heat transfer or woven FROG label, 63 millimeters x 63 millimeters (2.480 inches x 2.480 inches) in size adhered or sewn to the lower left outside of the undershirt on the front (as worn), approximately 1 inch above the hem. If a woven label, it shall be a cut single, fused edge label, manufactured by ITW Graphics, Bell Label, or equal. The label shall be in the following configuration:



3.3.2.2.3 Combination size, identification, and instruction label. Each undershirt shall have a heat transfer combination size, identification, and instruction label manufactured by ITW Graphics, Bell Label, or equal. The combination size, identification, and instruction label shall be adhered to the inside of the undershirt at the center back neck ¾±¼ inch below the collar joining seam. Placement of the combination size, identification, and instruction label shall be to the right (as worn) of the USMC exclusive label. The heat transfer label shall contain the following information and be in the following configuration:

Size and identification information:

UNDERSHIRT, FR LIGHTWEIGHT

CONTRACT NO:

NSN:

FIBER CONTENT:

CONTRACTOR'S NAME:

SIZE:

Instruction Information:

Machine wash in lukewarm water (105 °F)

Tumble dry low or line dry

Do Not Use Fabric Softeners

DO NOT BLEACH, DRY CLEAN, OR IRON

3.3.2.2.4 Barcode label/tag. Each undershirt shall be individually barcoded with a label/tag conforming to type VI, class 17 of MIL-DTL-32075 when tested as specified in 4.4.1. The label/tag shall be located so that it is completely visible on the item when folded or packaged as specified (see 6.2) and shall cause no damage to the item.

3.4 Design. The undershirt shall have a rib knit mock turtleneck and cuffs, offset shoulder seams and long sleeves with an underarm gusset. All joining seams shall be a flat locked construction to reduce chaffing and provide comfort in movement.

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3.4.1 Figures. Figures 1 and 2 are furnished for informational purposes only. To the extent an inconsistency exists between the written specification and the figures, the written specification shall govern.

3.5 Patterns. Standard patterns, providing a seam allowance of ⅜ inch for all seams except where otherwise specified, will be furnished by the Government (see 6.2). The list of pattern parts and computer nomenclature in 3.5.1 is provided to ensure that the pattern set provided is complete. The Government patterns shall not be altered in any way, and shall be used only for cutting the contractor's working patterns. The working patterns shall be identical to the Government patterns, except that additional notching, if needed to facilitate manufacture is permitted on the contractor's working pattern. Minor modifications are permitted where necessary to accommodate the manufacturer's processes and the use of automated equipment provided the modifications do not alter the dimensional, serviceability, or appearance requirements cited in this document or governed by the Government pattern.

3.5.1 Pattern parts. Component parts shall be cut from the materials specified and in accordance with the pattern parts listed in table II.

TABLE II. List of pattern parts.

Material	Piece Name	Cut Parts
Basic material	Back	1
	Front	1
	Shoulder	2
	Underarm gusset	2
	Sleeve	2
Cuff material	Collar	1
	Cuff	2

3.6 Construction. The following specifics will provide garments with uniform appearance, comfort, and durability for use during field duties, combat, and operations other than war. End item construction and appearance shall conform to the requirements and figures provided and the finished dimensions cited (see 3.7) to maintain end item configuration.

3.6.1 Seams and stitching. The seams and stitching used shall be consistent, exhibit a uniform appearance, and conform to ASTM D6193 and the requirements for the location, stitch, and seam types listed in table III. A seam allowance of ⅜-inch gauge is provided for all seaming. Seam allowances shall be maintained with seams sewn so that no raw edges, runoffs, twists, pleats, puckers, or open seams result. All seams shall start and finish evenly and edges shall not unravel. Thread tension shall be maintained so that there is no tight, loose, or unbalanced stitching. The stitches per inch shall be 10-14 for all seams except stitch type 607 which shall be 12-14 stitches per inch.

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TABLE III. Seams and stitching.

Seaming Area and Operations	Stitch Type	Seam Type
Join shoulders to front and back	607	FSa-1
Fold cuff in half and join to sleeve	607	FSa-1
Set sleeves flat	607	FSa-1
Join gusset to back and sleeve	607	FSa-1
Close sleeves and cuffs	607	FSa-1
Join collar edges	504	SSa-1
Set collar with seam at center back	504	SSa-1
Turn collar down and cover stitch, catching seam	406	SSh-2
Turn under undershirt hem ¾±⅛ inch and cover stitch	406	EFa-2
Turn under seam tails at cuff ends and secure with ⅝-inch bartack	Bartack	N/A
Secure hem ending with a ⅝-inch bartack	Bartack	N/A
Place combination instruction/ID label at center back ¾±¼ inch below the collar opening and apply with heat transfer machine or stitch to back around all four sides	N/A	N/A
Place FROG label on lower left front as worn approximately 1 inch above hem and apply with heat transfer machine	N/A	N/A

3.6.2 Bartacks. Bartacks for reinforcement shall be placed in the length, quantity, and location as listed in table IV.

TABLE IV. Bartacks.

Bartacks Locations	Length (Inches)	Quantity per Garment
Bottom of cuff at underarm seam	⅝	2
Undershirt hem at side seam	⅝	1

3.6.3 Tacking and backstitching. Ends of seams and rows of stitching, when not caught in other seams or stitching, shall be securely backstitched not less than ¼ inch. Thread breaks (all stitch types) shall be secured by stitching back of the break not less than ½ inch.

3.7 Finished dimensions. The finished undershirt shall conform to the measurements in table V as defined in table VI. All measurements shall be taken with the garment laid flat. Sleeve lengths shall not be uneven by more than ½ inch.

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TABLE V. Finished measurements (inches).

	X-Small	Small	Medium	Large	X-Large	XX-Large	XXX-Large	Tolerance
½ chest	20½	22	23½	25	26½	28	29½	±1
Back length	29	29½	30	30½	31	31½	32	±½
Sleeve outseam	24⅛	24⅜	24⅝	24⅞	25⅛	25⅜	25⅝	+½ -¼
½ cuff opening	3	3¼	3½	3¾	4	4¼	4½	±¼
½ neck opening	8	8⅛	8¼	8½	8¾	9	9½	±¼

TABLE VI. Methods of measure.

Measurement	Method
½ chest	Lay undershirt on a flat surface with sleeves laid straight up above shirt, i.e. gusset lays flat. Measure from folded edge to folded edge at base of sleeve setting seam.
Back length	Measure from center of back at neck seam to bottom of shirt hem.
Sleeve outseam	Measure in a straight line from shoulder seam at the outer folded edge to base of cuff.
½ cuff opening	Measure from folded edge to folded edge at base of finished cuff.
½ neck opening	Lay undershirt on a flat surface, align top edges of collar, and measure from folded edge to folded edge along collar joining seam.

3.8 Toxicity. The undershirt shall not present a health hazard and shall show compatibility with prolonged direct skin contact when tested as specified in 4.4.5. Chemicals recognized by the Environmental Protection Agency (EPA) as human carcinogens shall not be used. When specified (see 6.2), the contractor shall furnish information, which certifies that the finished product is composed of materials, which have been safely used commercially or provide sufficient toxicity data to show compatibility with prolonged, direct skin contact.

3.8.1 Toxicity documents. As specified (see 6.2), all finishes/chemicals used to process the garment shall be identified and accompanied by the appropriate Safety Data Sheet (SDS) information.

3.9 Workmanship. The finished undershirt shall be free from loose thread, foreign matter, and irregular defects that can adversely affect usage or durability. The finished undershirts shall conform to the quality of product established by this document, be uniform, and be free from defects specified in 4.4.3 and those defects that adversely affect form, fit, or function. Defects shall not exceed the established Acceptable Quality Levels (AQLs) (see 6.2) and shall not adversely affect the serviceability, appearance, and uniformity of the product.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. The first article, in accordance with 3.1.1, shall be examined in accordance with 4.4.1 through 4.4.5 for compliance with design, configuration, workmanship, dimensional requirements, and testing as specified in this document (see 6.2).

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4.2.1 First article samples and acceptance criteria. Unless otherwise specified in the contract (see 6.2), first article samples shall be provided. The sample size shall be as specified (see 6.2). The sample unit shall be one undershirt and the lot size shall be expressed in units of undershirts. The presence of defects exceeding AQLs (see 6.2) or failure of any testing shall be cause for rejection of the first article.

4.3 Conformance inspection. Unless otherwise specified in the contract (see 6.2), conformance inspection, in accordance with 3.1.2, shall consist of the examinations and tests as specified in 4.4.1 through 4.4.5.

4.3.1 Material and component sampling and acceptance criteria. Unless otherwise specified in subsidiary specifications, referenced documents, or procurement documents (see 6.2), sampling shall be in accordance with table VII. The lot shall be unacceptable if one or more sample units fail to meet any test requirement specified. For material testing, the sample unit shall be 5 continuous yards full width of the finished cloth for all physical and chemical tests.

TABLE VII. Material sampling.

Lot Size (Yards)	Sample Size (Sample Unit)
800 or less	2
801 through 22,000	3
22,001 and over	5

4.3.2 End item sampling and acceptance criteria. Unless otherwise specified in the contract (see 6.2), sampling for inspection shall be performed conforming to ANSI/ASQ Z1.4. The sample unit shall be one finished undershirt and the lot size shall be expressed in units of undershirts. The presence of excessive defects (see 6.2) or failure of any testing as specified (see 6.2) shall be cause for rejection of the lot.

4.4 Inspection methods.

4.4.1 Component and material examinations and tests. In accordance with 4.1, components and materials shall be inspected and tested in accordance with all the requirements of this specification and those referenced specifications and standards unless otherwise excluded, amended, or modified in this specification or applicable procurement documents (see 6.2). In addition to testing provisions specified in referenced documents, materials and components shall be examined and tested in accordance with table VIII and table IX. All requirements are applicable to the sample unit.

TABLE VIII. Component and material verification.

Material	Requirement	Verification
Basic material	3.3.1.1	As listed in table IX
Cuff material	3.3.1.2	As listed in table IX
Thread	3.3.2.1	A-A-55217
Barcode label tag	3.3.2.2.4	MIL-DTL-32075

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TABLE IX. Material verification.

Material Characteristic	Material Requirement		Test Method
	Basic Material	Cuff Material	
Construction	3.3.1.1	3.3.1.2	Visual 1/
Fiber content	3.3.1.1	3.3.1.2	AATCC TM20A 1/
Weight (minimum)	3.3.1.1	3.3.1.2	ASTM D3776/D3776M (method C)
Colorfastness (minimum)			
Laundering (3 cycles)	3.3.1.1	3.3.1.2	AATCC TM61, option 2a
Crocking			
Wet	3.3.1.1	3.3.1.2	AATCC TM8
Dry	3.3.1.1	3.3.1.2	AATCC TM8
Light, Xenon	3.3.1.1	3.3.1.2	AATCC TM16.3, option 2 (85 kJ)
Dimensional stability (3 cycles) (%) (maximum)	3.3.1.1	3.3.1.2	AATCC TM135, (1), III, (A), ii
Wales and courses	3.3.1.1	3.3.1.2	
Bursting strength (minimum)	3.3.1.1	3.3.1.2	ASTM D3787
Pilling (minimum)	3.3.1.1	3.3.1.2	ASTM D3512/D3512M
Visual shade matching	3.3.1.1	3.3.1.2	
Daylight (D75) (7500±200K)	3.3.1.1	3.3.1.2	AATCC EP9, option A
Incandescent (2856±200K)	3.3.1.1	3.3.1.2	AATCC EP9, option A
Air permeability (minimum)	3.3.1.1	3.3.1.2	ASTM D737
Fabric stretch (length) (minimum)	3.3.1.1	3.3.1.2	ASTM D2594 (Loose Fit)
Fabric growth (length) (maximum)	3.3.1.1	3.3.1.2	ASTM D2594 (Loose Fit)
Water sorption and wicking			
Back (water sorption)	3.3.1.1	3.3.1.2	4.4.1.1
Face (wicking) (minimum)	3.3.1.1	3.3.1.2	4.4.1.1
MVTR (grams/square meter/24 hours) (minimum)	3.3.1.1	3.3.1.2	ASTM E96/E96M 2/
Moisture wicking (inches)	3.3.1.1	3.3.1.2	4.4.1.2
Wales and courses	3.3.1.1	3.3.1.2	
Drying time (minutes) (maximum)	3.3.1.1	3.3.1.2	4.4.1.3

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TABLE IX. Material verification – Continued.

Material Characteristic	Material Requirement		Test Method
	Basic Material	Cuff Material	
Vertical flame (initial) Wales and courses (maximum)			ASTM D6413/D6413M
After flame, seconds	3.3.1.1	3.3.1.2	
After glow, seconds	3.3.1.1	3.3.1.2	
Char length, inches	3.3.1.1	3.3.1.2	
Melt/drip	3.3.1.1	3.3.1.2	
Vertical flame (after 5 launderings) Wales and courses (maximum)			AATCC TM135, 5 cycles, (1), III, (A), ii ASTM D6413/D6413M
After flame, seconds	3.3.1.1	3.3.1.2	
After glow, seconds	3.3.1.1	3.3.1.1	
Char length, inches	3.3.1.1	3.3.1.1	
Melt/drip	3.3.1.1	3.3.1.1	
NOTES: 1/ Results recorded as pass or fail. 2/ ASTM E96/E96M performed according to Procedure BW-Inverted Water Method.			

4.4.1.1 Water sorption and wicking. Water sorption and wicking shall be determined using the following procedure.

4.4.1.1.1 Water sorption and wicking test specimens. Fabric specimens shall be conditioned in accordance with ASTM D1776/D1776M and tested in that environment. The specimen size shall be 6 inches by 6 inches; three (3) separate specimens shall be used for each of the face side and back side tests. A fabric shall be considered passing only when tests on both the face side and the back side meet the respective test pass/fail criteria on all individual specimens tested.

4.4.1.1.2 Face side wicking test. The test specimen shall be laid flat on a glass plate with back side up (i.e., inner or skin surface when used in a garment). One (1) drop of 0.10±0.01 milliliter of distilled water at 70±2 °F shall be placed in the center of the test specimen using a pipettor and a stopwatch/timer immediately started. The test specimen shall then be immediately turned over on the glass plate with test specimen face side up. The diameter of the wicked water area (denoted by a darkened water mark) shall be measured at a total elapsed time of 10 seconds. The specimen shall be considered passing if the diameter of the wicked water area (darkened water mark) is equal to or greater than 0.4 inch.

4.4.1.1.3 Back side sorption test. The test specimen shall be laid flat on a glass plate with back side up (i.e., inner or skin surface when used in a garment). One (1) drop of 0.10±0.01 milliliter of distilled water at 70±2 °F shall be placed in the center of the test specimen using a pipettor and a stopwatch/timer immediately started. The water (denoted as a darkened water mark) shall be observed and the time for the water mark to disappear (water sorption, denoted as a lightened water mark approximating the shade of the basic material) shall be recorded. The specimen shall be considered passing if the water sorption (disappearance of the darkened water mark) is 5 seconds or less.

4.4.1.2 Wicking. Wicking data shall be collected using the following procedure:

a. A sample size of 6 inches by 1 inch shall be cut with the 6-inch cut measured in the Wale direction.

b. A 500-milliliter Erlenmeyer flask shall be filled with 200 milliliters of colored water (food coloring used shall be a contrasting color to the cloth material to make the water level visible on the sample).

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c. The top edge of the sample shall be pierced with a long straight pin and the sample then suspended from the top of the flask.

d. After 1 minute, the sample shall be removed from the flask and the water level on the sample measured in inches and recorded.

e. The sample shall be returned to the flask.

f. The following measurement shall be repeated at 3 minutes, 5 minutes, and each 5-minute interval until the water level reaches 6 inches or 1 hour has elapsed. Values are recorded as inches per hour. A minimum of 3 samples shall be tested with the average value recorded along with each individual measurement.

4.4.1.3 Drying time test method.

4.4.1.3.1 Apparatus and materials. The following apparatus and materials shall be used:

a. Wringer (motor driven) (see 4.4.1.3.4.a).

b. Laboratory balance, accurate to 0.01 gram.

c. White AATCC Textile Blotting Paper, 25 x 25 centimeters (see 4.4.1.3.4.b).

d. Water, distilled.

e. 250-milliliter glass beaker.

4.4.1.3.2 Test specimens. The fabric samples and blotting paper should be conditioned at 65±2 percent RH and 70±2 °F for a minimum of 4 hours. Three (3) 2- by 2-inch test specimens shall be tested per sample.

4.4.1.3.3 Drying time test procedure. The following drying time test procedure shall be followed:

a. Run test in standard conditions: 65±2 percent RH and 70±2 °F.

b. Weigh the conditioned specimen using a laboratory balance accurate to 0.01 gram.

c. Place 100 milliliters of distilled water into a 250-milliliter glass beaker.

d. Submerge the specimen in the beaker of water for 30 minutes. Make certain that the specimen is completely submerged to insure complete wetting.

e. Remove the specimen and sandwich it between two pieces of unused blotting paper. Pass the sandwich through the wringer with a dead weight load of 27.7±0.5 kilograms.

f. Immediately place specimen on the balance with top door of the balance open, side doors closed and record wet weight either to the nearest 0.01 or 0.1 gram. (Degree of dryness shall be determined by the user.) Manually monitor weight at set intervals until dry or use an automated balance with capability to weigh specimen until dry (see 4.4.1.3.4.c). Record time to dry.

g. Repeat for remaining specimens. Average the 3 specimens.

4.4.1.3.4 Drying time test notes.

a. Wringer (motor driven): see AATCC TM70, footnote 11.2.

b. Blotting paper: see AATCC TM70, footnote 11.3.

c. Balance: Suitable Automation Software, PerkinElmer Inc., http://www.perkinelmer.com/.

4.4.2 In-process examination. Visual and dimensional examinations shall be made at any point or during any phase of the manufacturing process to determine whether construction details, which cannot be examined in the finished product, are in accordance with the requirements specified in section 3. Materials and components which contain defects and damages, as defined in the visual examination of 4.4.3, shall be removed from production. Any in-process nonconformance remaining in the finished undershirt shall be classified as a defect in accordance with the visual examination of 4.4.3.

4.4.3 Visual examination. The undershirts shall be examined for defects in color, shade, design, material, construction, workmanship, and marking, and the defects shall be classified in accordance with the list of defects specified in table X. The inspection conditions and the AQLs shall be as specified (see 6.2).

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TABLE X. Visual examination.

Part	Defects	Classification	
		Major	Minor
Material	a. Hole, cut, tear, smash, burn, drill hole, run, needle chew, visible mend, thin place, misweave affecting appearance or serviceability.	101	
Material components and assembly	a. Component part omitted, distorted, full, tight, or twisted; any part of the shirt caught in any unrelated stitching, the edge of any component part required to be forced out having folds of more than ⅛ inch.	102	
Shade	a. Shade variation within a part or between parts. Thread color not as specified.	103	
	b. Dye streak, color not as specified.	104	
Stitching and seams	a. Seams puckered, distorted, pleated, wavy, twisted; edge or raised stitching sewn too close to edge, resulting in damage to cloth.		201
	b. Seams irregular or open seam, raw edge affecting appearance or serviceability.	105	
	c. Seam, seam allowance, or stitch type not as specified.		202
	d. Loose or tight stitch tension; broken or missing thread or stitch		203
Bartacks	Bartacks missing, insecure, misplaced, not specified size, stitches loose or broken.		204
Cleanliness	Spot, stain, excessive thread ends not trimmed or removed, odor.		205
Labels	a. Label omitted, incorrect type, size or configuration; illegible, inaccurate, improperly attached or peeling.	106	
	b. Barcodes not readable by scanner; human-readable interpretation (HRI) omitted or illegible.		206
	c. Barcode not visible on folded, packaged item.		207
	d. Barcode attachment causes damage to the item.	107	
Packaging	Any undershirt not packaged in accordance with the contract or purchase order.		208

4.4.4 Dimensional examination. The undershirt shall be examined for conformance to the dimensions specified in table V and the methods of measure specified in table VI. Any measurement deviating from dimensions and tolerances specified in table V as defined in table VI shall be scored as a measurement defect. The inspection conditions and the AQL shall be as specified (see 6.2).

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4.4.5 Toxicity assessment. An acute dermal irritation study and a skin sensitization study shall be conducted on laboratory animals. When the results of the studies indicate the undershirt is not a sensitizer or irritant, a Repeat Insult Patch Test shall be performed in accordance with the Modified Draize Procedure (Hayes' Principles and Methods of Toxicology and Contact Allergy: Predictive Testing in Humans). If the toxicity requirement (see 3.8) can be demonstrated with historical data, then toxicity testing may not be required. At a minimum, toxicity data should include results from a primary dermal irritation study in laboratory animals and a repeated insult human patch test (Modified Draize Procedure). The latter must be conducted under the supervision of a qualified dermatologist using at least 100 free-living individuals.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The undershirt is part of the underwear set intended for use with the cold weather clothing system and as part of the FROG worn by personnel of the USMC.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Sizes required (see 1.1)

c. National stock numbers (NSNs).

d. When first article is required (see 3.1.1, 4.2, and 6.3).

e. When conformance inspection is required (see 3.1.2).

f. Applicable Government patterns, drawings, and reference standards, including revisions (see 3.3.1.1, 3.3.1.2, and 3.5).

g. Requirement for the submittal of information which certifies that the finished product is composed of materials which have been safely used commercially; or sufficient toxicity data to show compatibility with prolonged, direct skin contact (see 3.8).

h. Requirement for an SDS and the identification of all finishes/chemicals used to process the garment (see 3.8.1).

i. AQLs (see 3.9, 4.4.3, and 4.4.4).

j. First article sampling and acceptance criteria (see 4.2.1).

k. Conformance inspection (see 4.3).

l. Component and material sampling and acceptance criteria (see 4.3.1).

m. End item sampling and acceptance criteria (see 4.3.2).

n. Component and material examination and tests (see 4.4.1).

o. Inspection conditions (see 4.4.34.4).

p. Packaging requirements (5.1).

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6.3 First article. When first article inspection (see 4.2) is required, the contracting officer should provide specific guidance to offerors whether the item(s) should be first article sample, a first production item, or a standard production item from the contractor's current inventory, and the number of items to be tested as specified in 4.2.1. Invitations for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for pending contract. Bidders should not alternate submit bids unless specifically requested to do so in the solicitation.

6.4 Suggested sources.

a. Basic and cuff fabric:

(1) Longworth Industries, Inc., 575 Air Tools Drive, Suite K, Southern Pines, NC 28387

b. Woven and adhesive labels:

(1) Bell Label, 777 Main St., Lewiston, ME 04240, (207) 784-2961

(2) ITW Graphics, 375 New State Rd., Manchester, CT 06042, (860) 533-5107

6.5 NSNs. For informational purposes only, an NSN that corresponds to this specification is listed in table XI. Table XI is not intended to be a comprehensive list of all NSNs associated with this specification.

TABLE XI. NSNs.

NSN	Size
8415015674423	Medium

6.6 Supersession data. This document supersedes MC/PD 05 2011 SYSCOM, dated 21 September 2011, in its entirety.

6.7 Subject term (key word) listing.

Level 1

Marine Corps

Moisture wicking

Silkweight

Underwear

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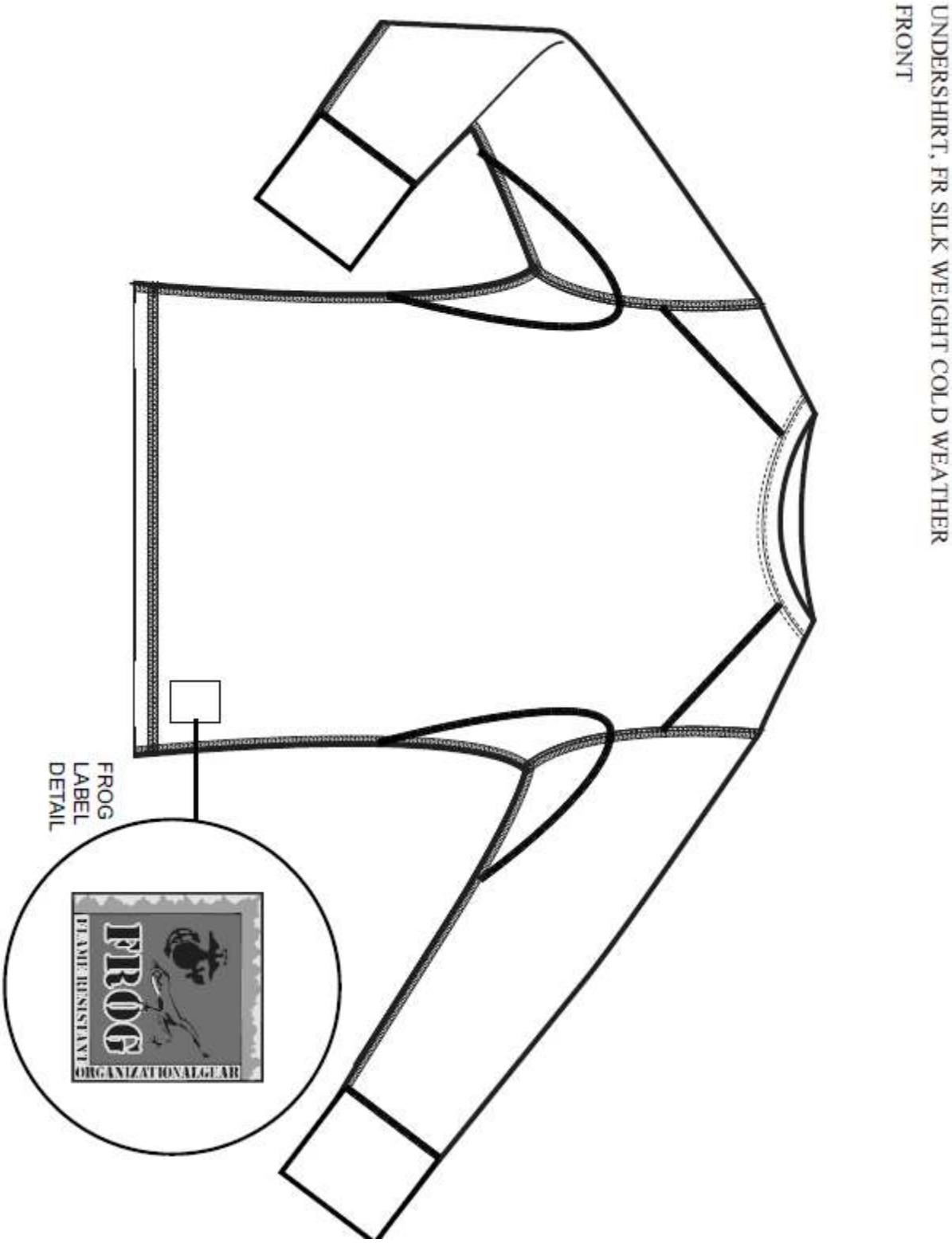


FIGURE 1. Front view.

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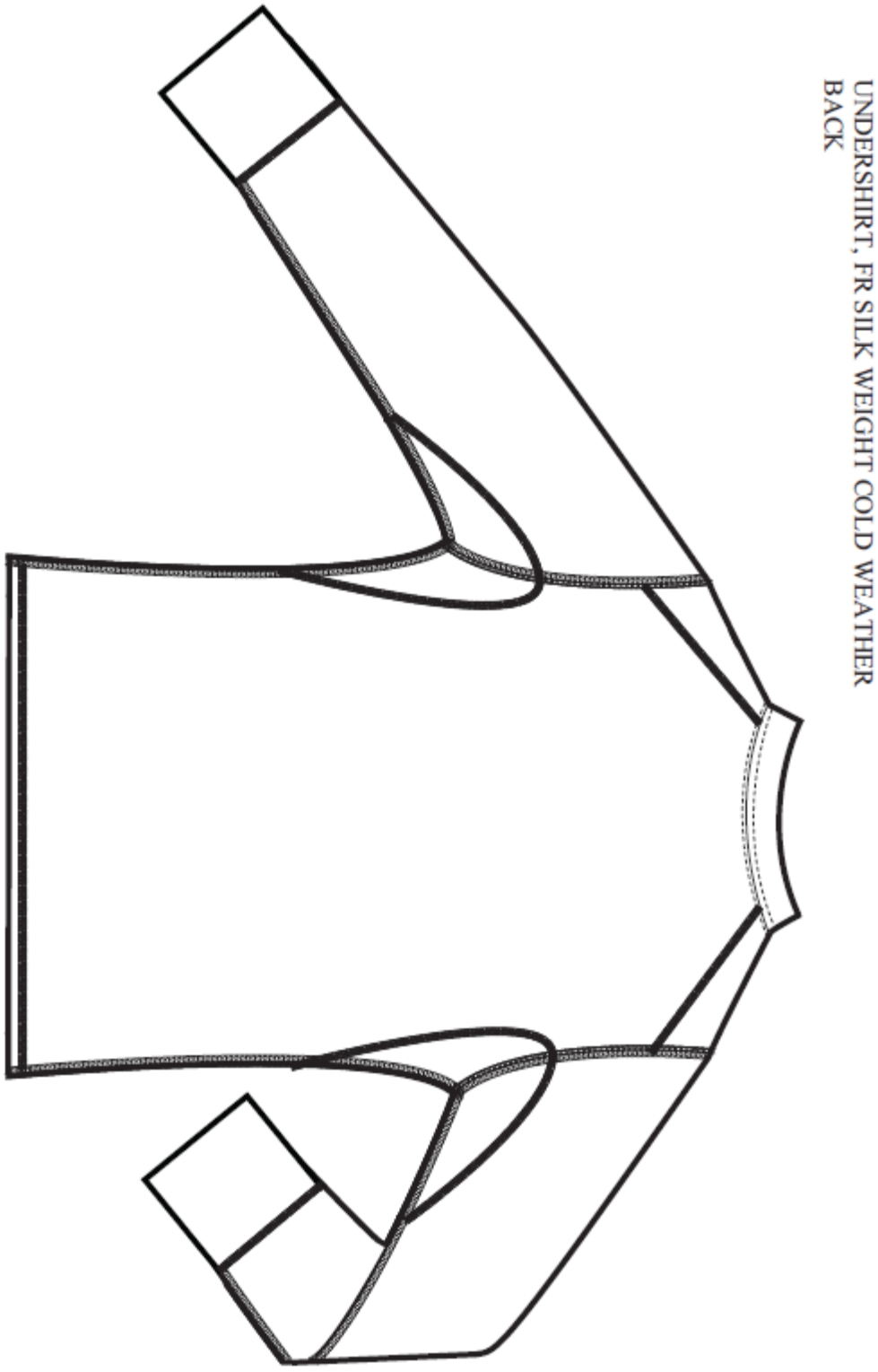


FIGURE 2. Back view.

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CONCLUDING MATERIAL

Custodians:
- Army – GL
- Navy – MC
- Air Force – 11

Review activities:
- Navy – CG1
- Air Force – 03, 70
- DLA – CT

Civil agency:
- GSA – FAS

Preparing activity:
- Navy – MC
- (Project 8415-2018-001)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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